



Semester One Examinations 2023 / 2024

Course Instance 4BCT1, 4BSE1, 1MSME1, 1MEEE1, 1MECS1, 1CSD1, 1CSD2,
Code(s) 2SPE1, 1MAI1, 1MEES1
Exam(s) B.Sc. Degree (Computer Science and Information
Technology) B.E. Energy Systems Engineering, MSc in
Mechanical Engineering, ME (Electrical and Computer
Engineering), ME (Electrical and Electronic Engineering),
M.Sc. in Computer Science (Data Analytics), M.Sc. in
Computer Science (Artificial Intelligence)

Module Code(s) CT561
Module(s) Systems Modelling and Simulation

Paper No. I

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Examiner(s)

Internal Prof. Michael Madden

Examiner(s) *Prof. Jim Duggan

Instructions: Answer 3 questions out of 4. All questions carry equal marks.

Duration 2 hrs

No. of Pages 5 (Including Cover Page)

Department(s) School of Computer Science

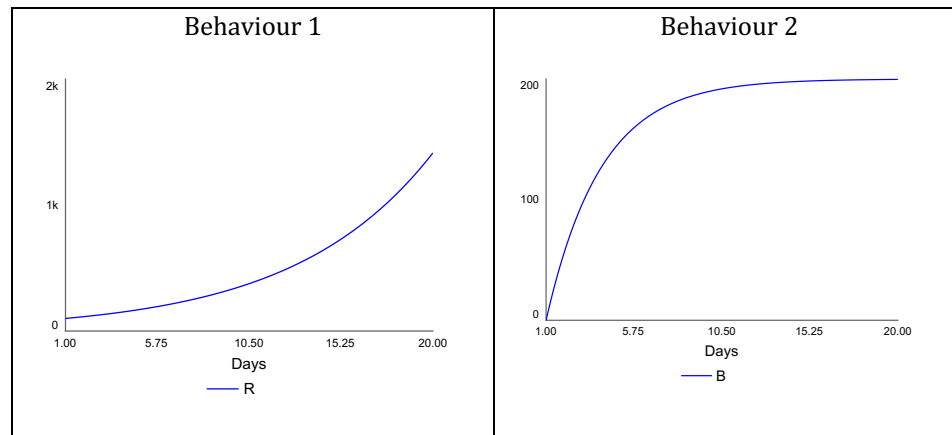
Course Directors Dr. Colm O'Riordan

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Requirements	Release in Exam Venue	Yes [X]	No []
	MCQ Answersheet	Yes []	No [X]
	Handout	None	
	Statistical/ Log Tables	None	
	Cambridge Tables	None	
	Graph Paper	YES	
	Log Graph Paper	None	
	Other Materials	None	
	Graphic material in colour	Yes [X]	No []

1. (a) Based on the two graphs (stocks) shown below, identify the most likely feedback structure that generates each individual plot.



Given that the models that generate these outputs are one-stock models, show separate stock and flow structures that would generate these behaviours.

[8]

- (b) Construct a stock and flow model from the following description for an order fulfillment company. Clearly show link polarities.

1. The company's *backlog* increases when *new orders* are placed, and reduces as *orders are fulfilled*.
2. As the *backlog* increases, so too does *work pressure*.
3. An increase in *work pressure* leads to a reduction in *time per task*.
4. An increase in *time per task* leads to a decrease in *orders fulfilled*.
5. An increase in *work pressure* leads to an increased *workweek*.
6. An increase in *workweek* leads to an increase in *orders fulfilled*.
7. An increase in *workweek* leads to an increase in *fatigue*.
8. An increase in *fatigue* leads to an increase in *time per task*.

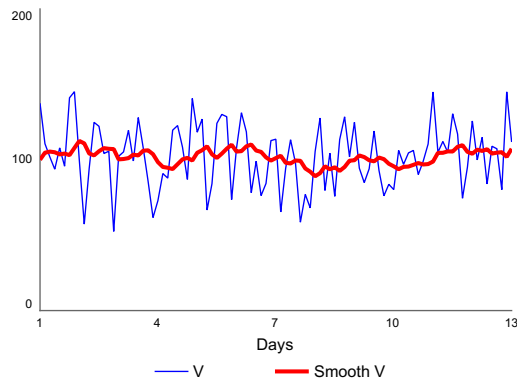
[10]

- (c) Based on the stock and flow model from part (b):

- Show the feedback loops and calculate their polarity, by tracing an increase in a variable all the way around a feedback loop, and observing its direction of change.
- Identify the feedback loop that can be viewed as a side-effect of increasing the workweek.

[7]

2. (a) This chart shows two variables, a random number stream (blue line) and a smoothed transformation of the random stream (red line).



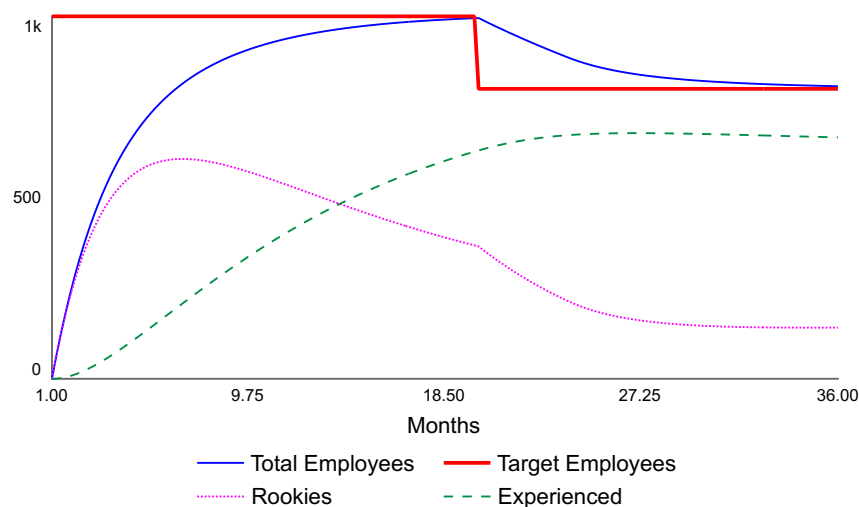
Show a stock and flow structure that could be used to generate the smoothed version, and discuss the role of the adjustment time parameter.

[8]

- (b) The stock management structure is composed of two elements: (1) the anchor and (2) the adjustment. Describe each of these elements, and specify what general type of model structure is used for each one.

[5]

- (c) The following graph is a simulation of staff (Rookies and Experienced) for a consulting company. Rookies progress to becoming experienced after an average duration of 12 months. The model also shows how the overall employee stock (Rookies + Experienced) is managed, as a change in the target results leads to changes in recruitment. Assume that people leave both Rookies (5%/month) and Experienced (2%/month). Create a stock and flow model that incorporates both an anchor and an adjustment. Specify all the equations, assume it models a startup, with no employees at the outset.



[12]

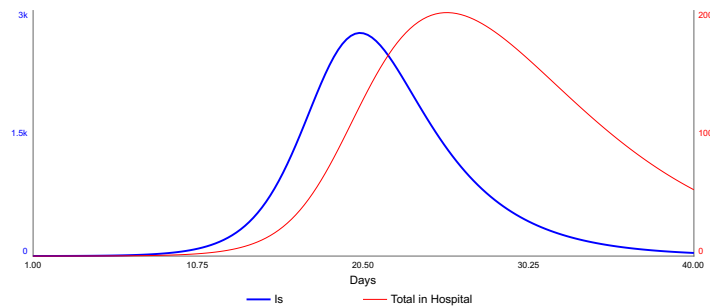
3. (a) Explain the formulation of these two flow equations, taken from the SIR model. I is the infected stock, c is average contacts per day, i is the chance that an encounter between infected will lead to an infection, S is the susceptible stock, N is the total population, and d is the average duration of infectiousness.

$IR = cI \left(\frac{S}{N} \right) i$	$RR = \frac{I}{d}$
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Define the equilibrium case for these two equations, and from that, show that $cid = 1$ is satisfied. For this, you may assume that $S = N$.

[10]

- (b) The following is an output from a SEI²H²R seven-stock Model. It is based on the SIR model structure, with a number of variations.

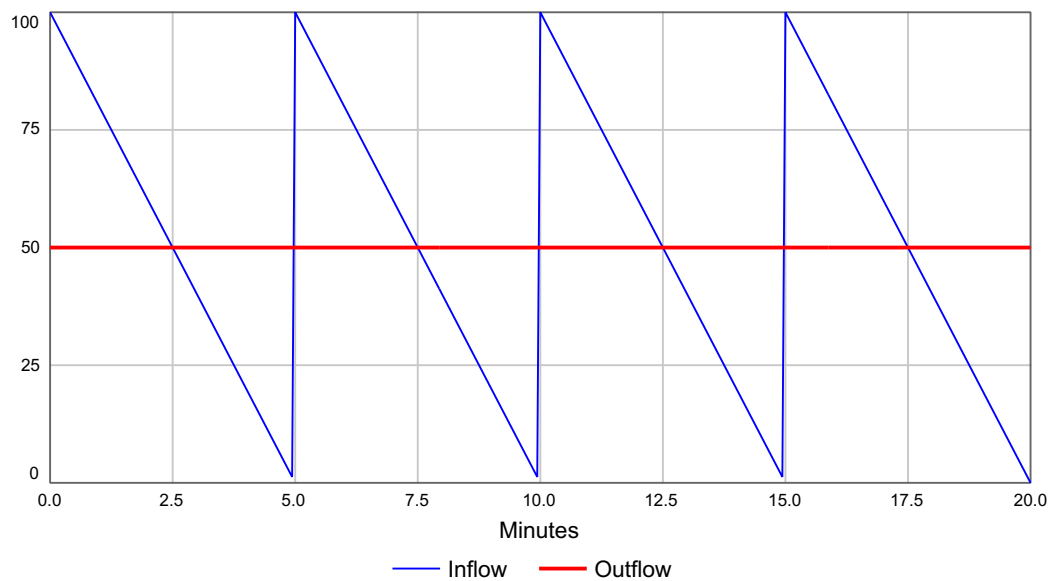


- There are two infectious stocks, infectious asymptomatic (I_a) and infectious symptomatic (I_s). These take their inflows from the Exposed (E) stock. The fraction symptomatic determines the number that flows to each of these compartments, and is set to 0.70.
- People in I_a flow directly into R , with a recovery delay of 4 days.
- A fraction ($1 - \text{Hospitalisation Fraction}$) of I_s flow into R after a recovery delay of 4 days, the remaining flow into the hospital stream. The hospitalisation fraction is set at 5%.
- The hospital stream is a second order delay structure of 10 days, which then flows into the recovered stock (R).
- The infection rate is based on the SIR model, but must separate the contribution from the stocks I_a and I_s . People in I_a are 50% less likely to transmit than those in I_s . The infection rate reduces Susceptibles (S) and increases Exposed (E).

Design a stock and flow model, with feedbacks, and show all the equations.

[15]

4. (a) Consider the flow variables shown in the chart below. The outflow (in red) is constant and equal to 50, while the inflow (in blue) varies in a regular pattern from 100 to 0 over five time units. Assuming that the initial value of the stock is 100, perform the following:
1. Draw a stock and flow model (including the net flow).
 2. Calculate and show the net flow.
 3. Use graphical integration to plot the stock over time, clearly explaining the different behaviour modes for each time interval.



[12]

- (b) Describe Euler's method of integration, and state the underlying assumptions.

Assuming that the net flow is linear, under what conditions will Euler's method overestimate the true value of a stock?

[5]

- (c) Based on the flows shown in part (a), calculate the stock using Euler's equation where $DT=5$.

[8]

END